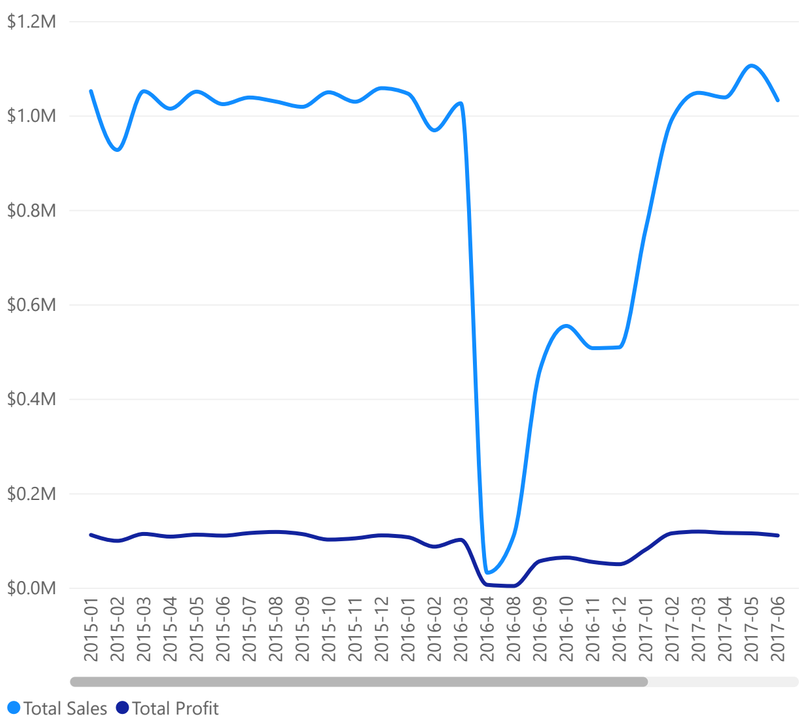
Supply Chain & Delivery Performance Command Center

Executive... | Delivery... | Profitability.. | Order... | Executive... | About &...

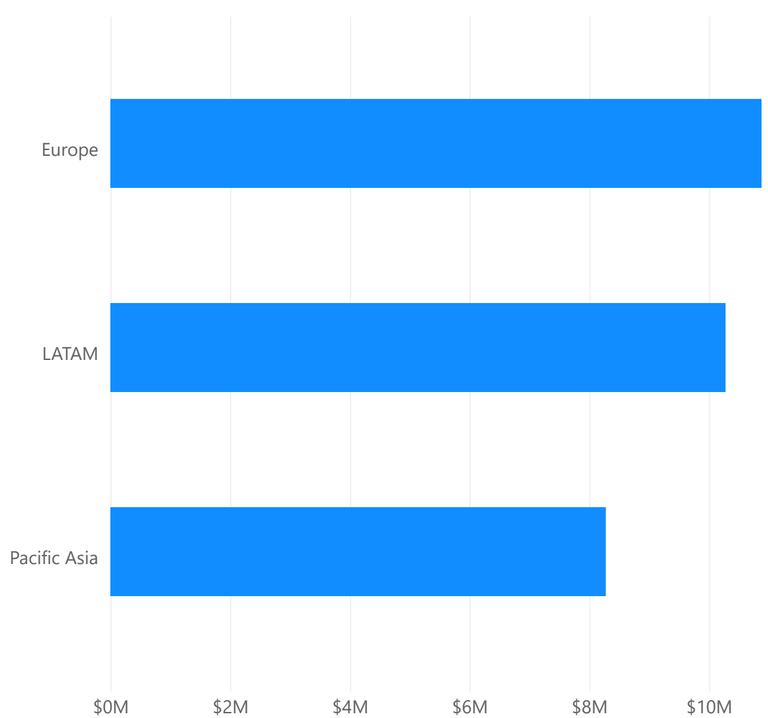
Year: All | Market: Multiple sele... | Shipping Mode: All

Total Sales	Total Profit	Profit Margin %	Total Orders	On-Time %
$29.42M	$3.15M	10.71%	53K	42.69%

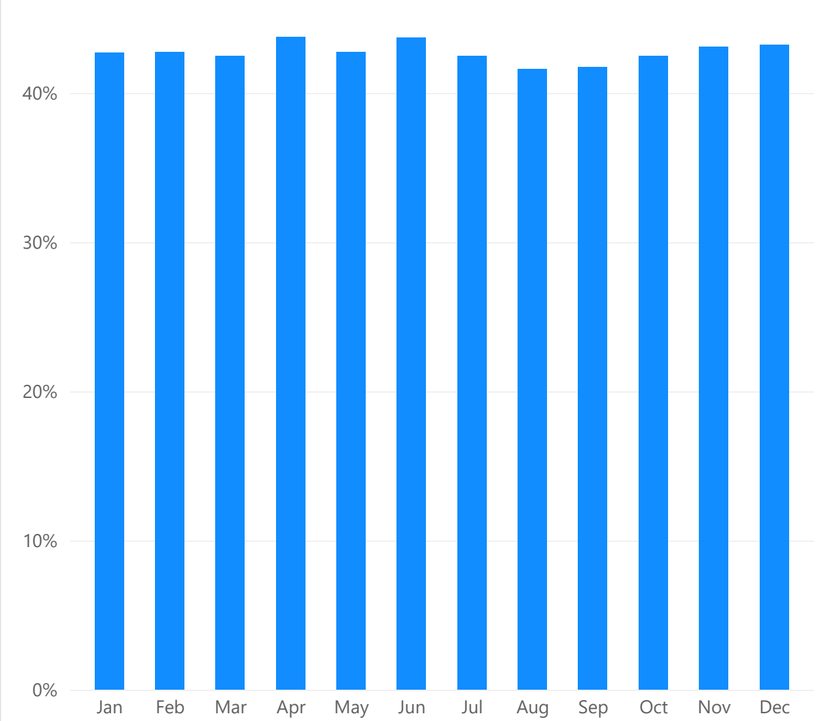
Total Sales and Total Profit
by Year and Month



Total Sales
by Market



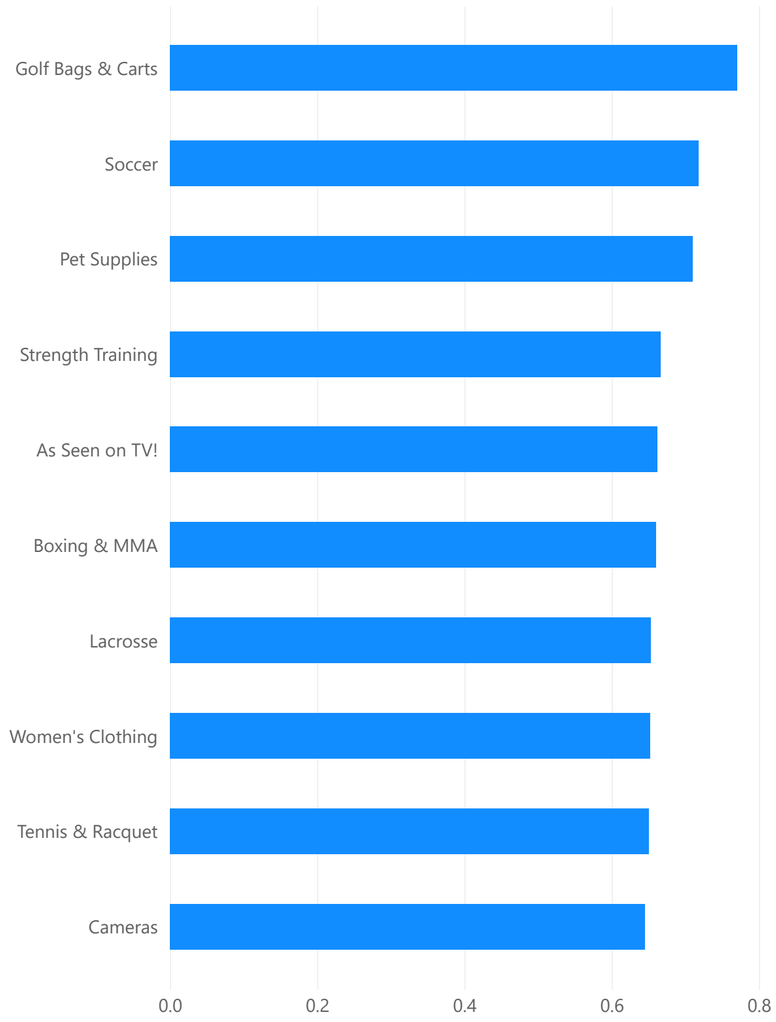
On-Time %
by Month Name



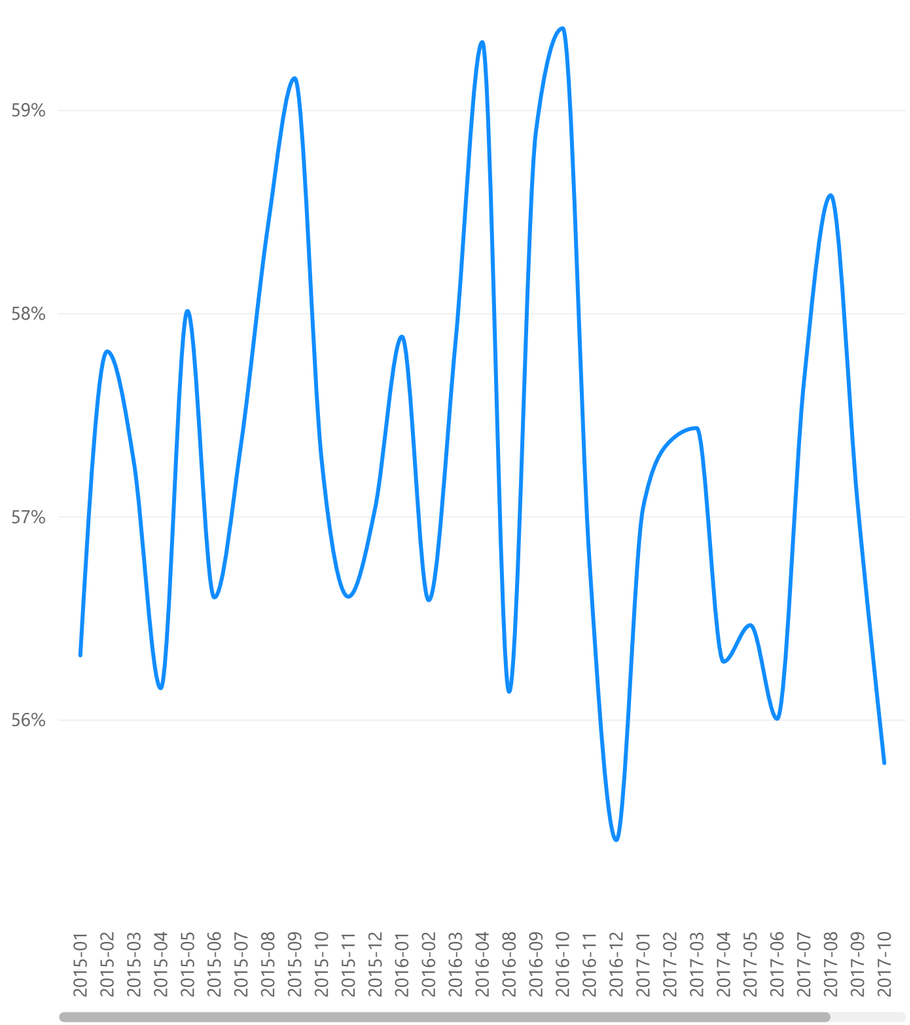
Delivery Performance

Year: All | Market: Multiple sele... | Shipping Mode: All

Avg Shipping Delay
by Category



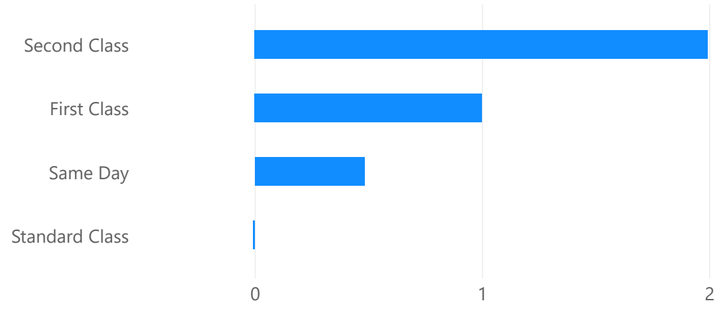
Late Delivery %
by Year Month



On-Time %
by Region & Shipping Mode

Order Region	Same Day	Second Class	Standard Class	**Total**
Caribbean	47.66%	17.51%	63.20%	**44.47%**
Central America	52.03%	21.70%	59.89%	**42.89%**
Central Asia	79.69%	9.82%	57.43%	**43.54%**
Eastern Asia	50.64%	16.78%	61.03%	**43.28%**
Eastern Europe	63.64%	23.44%	59.14%	**42.35%**
Northern Europe	58.96%	20.76%	60.54%	**43.75%**
Oceania	56.96%	20.51%	59.45%	**43.68%**
South America	46.43%	21.41%	59.97%	**42.82%**
South Asia	48.77%	20.19%	60.34%	**41.65%**
Southeast Asia	56.60%	20.42%	59.17%	**42.02%**
Southern Europe	43.36%	20.67%	61.67%	**43.20%**
West Asia	53.96%	20.40%	60.58%	**42.19%**
Total	**51.65%**	**20.14%**	**60.24%**	**42.69%**

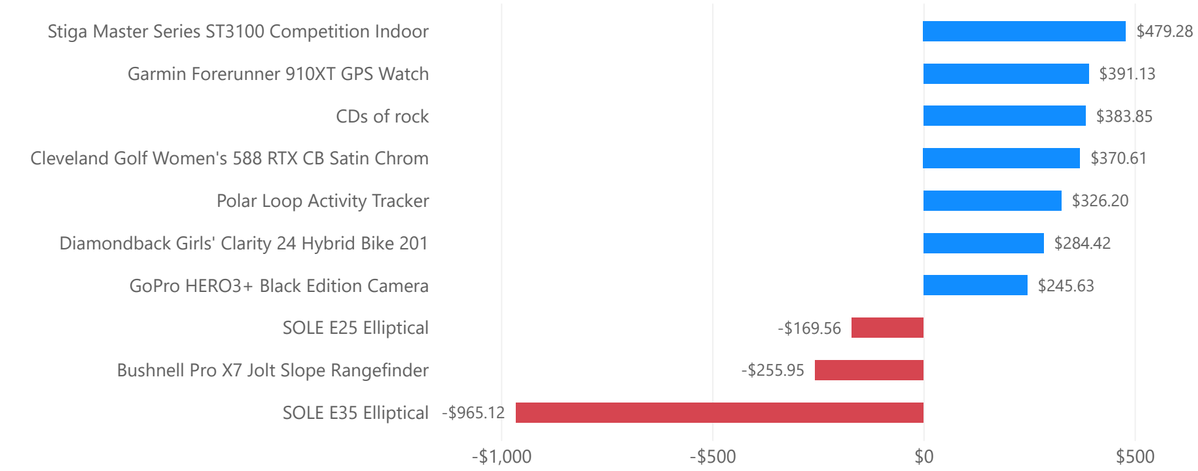
Avg Shipping Delay (Days)
by Shipping Mode



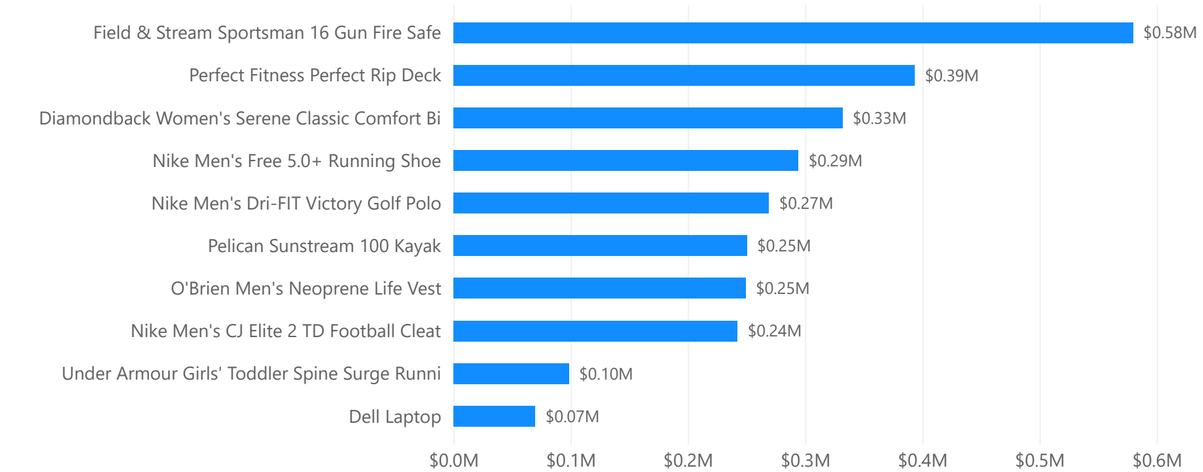
Profitability & Discount Impact Analysis

Year: All | Market: Multiple sele... | Shipping Mode: All

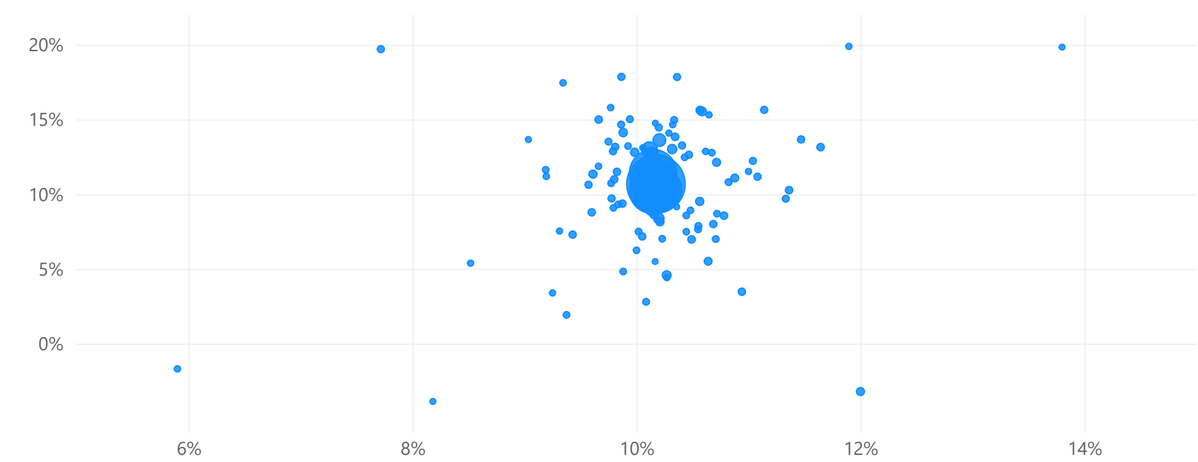
Least Profitable Products



Most Profitable Products

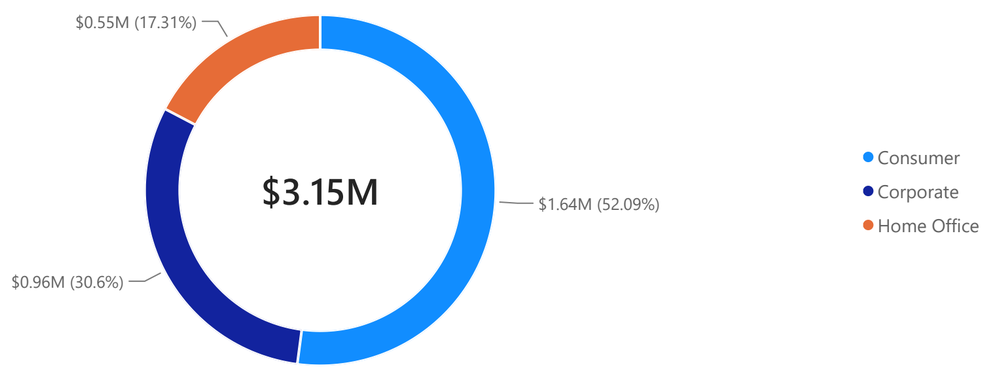


Discount Rate vs. Profit Margin



Profit Share

by Customer Segment



Executive Insights & Action Plan

Key Operational Findings

1. Shipping SLA Bottleneck
   - *Finding:* Second Class shipping represents a severe fulfillment failure, achieving an on-time delivery rate of only 20.14% and averaging ~2 full days of delay across all product categories.
   - *Visual Reference:* Delivery Performance -> *On-Time % by Region & Shipping Mode Matrix*.

2. Catalog Loss Concentration (Profitability Analysis Page)
   - *Finding:* Net losses across the catalog are extremely concentrated—90%+ of total net loss stems from just 3 items, led by the *SOLE E35 Elliptical* (-$965.12 net profit).
   - *Visual Reference:* Profitability Analysis -> *Bottom 10 Products by Profit Bar Chart*.

3. Discount Rate Margin Erosion (Profitability Analysis Page)
   - *Finding:* Order discount rates exceeding 10% show no statistically significant volume lift, but directly collapse profit margins down to near-zero or negative territory.
   - *Visual Reference:* Profitability Analysis -> *Discount Rate vs. Profit Margin Scatter Plot*.

4. Core Revenue & Profit Driver (Executive Overview Page)
   - *Finding:* The Consumer customer segment is the primary financial pillar, contributing 52.09% ($1.64M) of total net profit across 53K completed orders.
   - *Visual Reference:* Executive Overview -> *Top KPI Cards* & Profitability Analysis -> *Profit Share Donut Chart*.

5. Regional SLA Disparity (Delivery Performance Page)
   - *Finding:* Regional fulfillment exhibits high variance—*Southern Europe* (43.20% on-time) and *South Asia* (41.65% on-time) lag significantly behind primary markets.
   - *Visual Reference:* Delivery Performance -> *On-Time % Matrix*.

Strategic Recommendations

- **Renegotiate Second Class Carrier SLAs:**
  - Transition high-priority orders away from Second Class shipping or renegotiate carrier contracts in low-performing regions (*Southern Europe*, *South Asia*).
- **Delist / Reprice Loss-Making Products:**
  - Halt automated discounting or renegotiate supplier procurement costs for the 3 negative-margin fitness items (*SOLE E35*, *SOLE E25*, *Bushnell Pro*) to eliminate bottom-line losses.
- **Enforce Automated Discount Guardrails:**
  - Implement mandatory managerial approval in order processing for any discount rate exceeding **10%** to safeguard overall margin percentages.

Project & Dataset Overview

Project & Dataset Overview:

- **Data Source:** Kaggle DataCo Supply Chain Dataset (**180,000+ order lines**, **$29.42M total sales**, **53K orders**).
- **Environment:** Power BI Desktop (Windows), Power Query (M Language), DAX.
- **Architecture:** 1 Fact table (Fact_Orders) connected to 5 Dimension tables (Dim_Customers, Dim_Products, Dim_Geography, Dim_ShippingMode, Dim_Date) in a Star Schema model.

Business Rules & Key Metric Dictionary

- **Total Sales**
  - **Logic:** SUM(Fact_Orders[Sales])
  - **Definition:** Gross revenue generated across all valid (non-canceled) order lines.
- **Total Profit**
  - **Logic:** SUM(Fact_Orders[Order Profit Per Order])
  - **Definition:** Net profit accumulated after product manufacturing costs and discounts.
- **Profit Margin %**
  - **Logic:** [Total Profit] / [Total Sales]
  - **Definition:** Percentage of gross sales revenue retained as net profit.
- **On-Time %**
  - **Logic:** On-Time Orders / Total Valid Deliveries
  - **Definition:** Share of orders delivered where Actual Shipping Days <= Scheduled Days (excludes canceled shipments).
- **Late Delivery %**
  - **Logic:** Late Orders / Total Valid Deliveries
  - **Definition:** Share of completed orders that exceeded their scheduled shipment window.
- **Avg Shipping Delay (Days)**
  - **Logic:** AVERAGE(Fact_Orders[Days for shipping] - Fact_Orders[Scheduled days])
  - **Definition:** Average number of days orders were delayed past promised carrier SLAs.